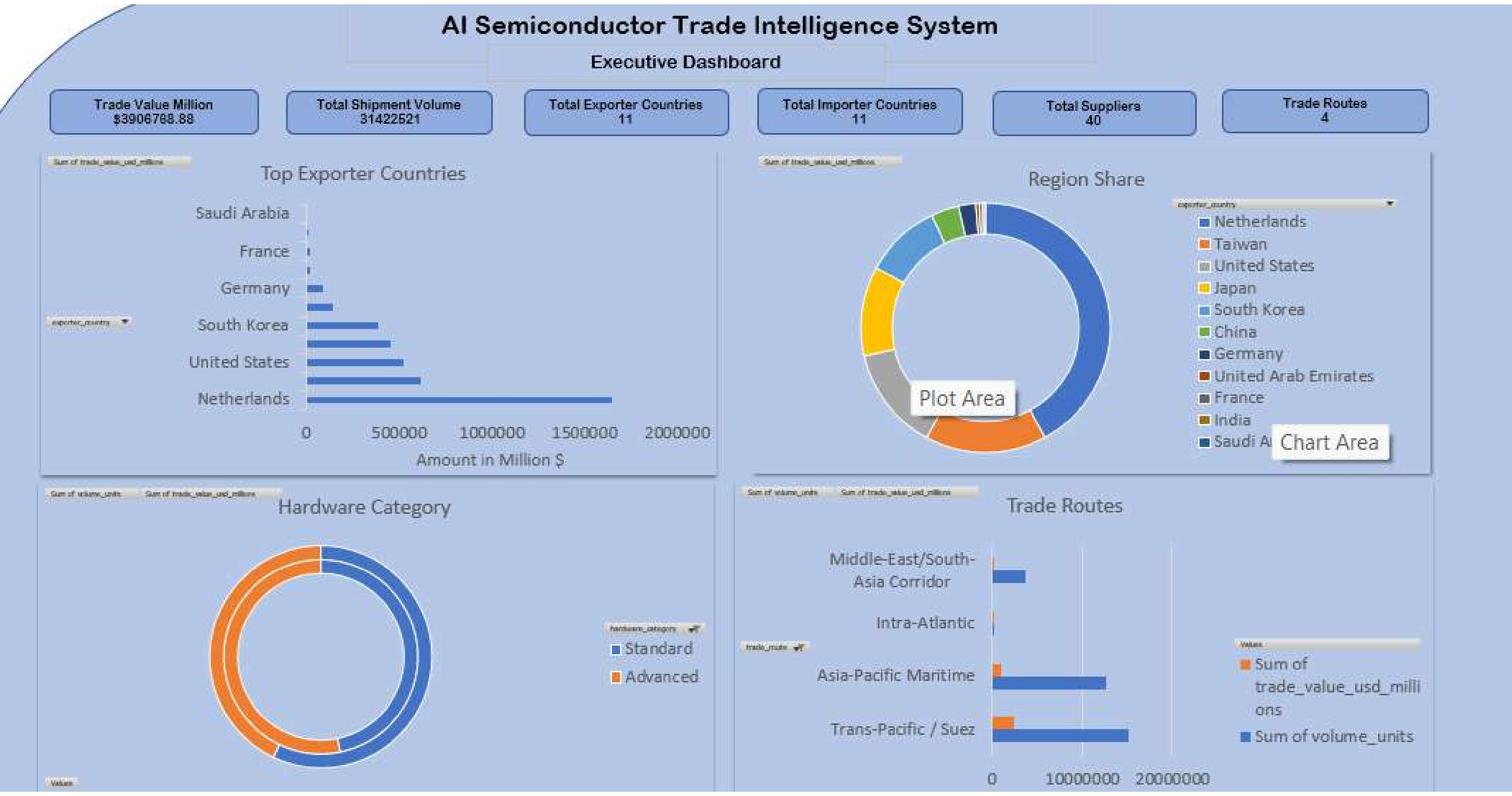
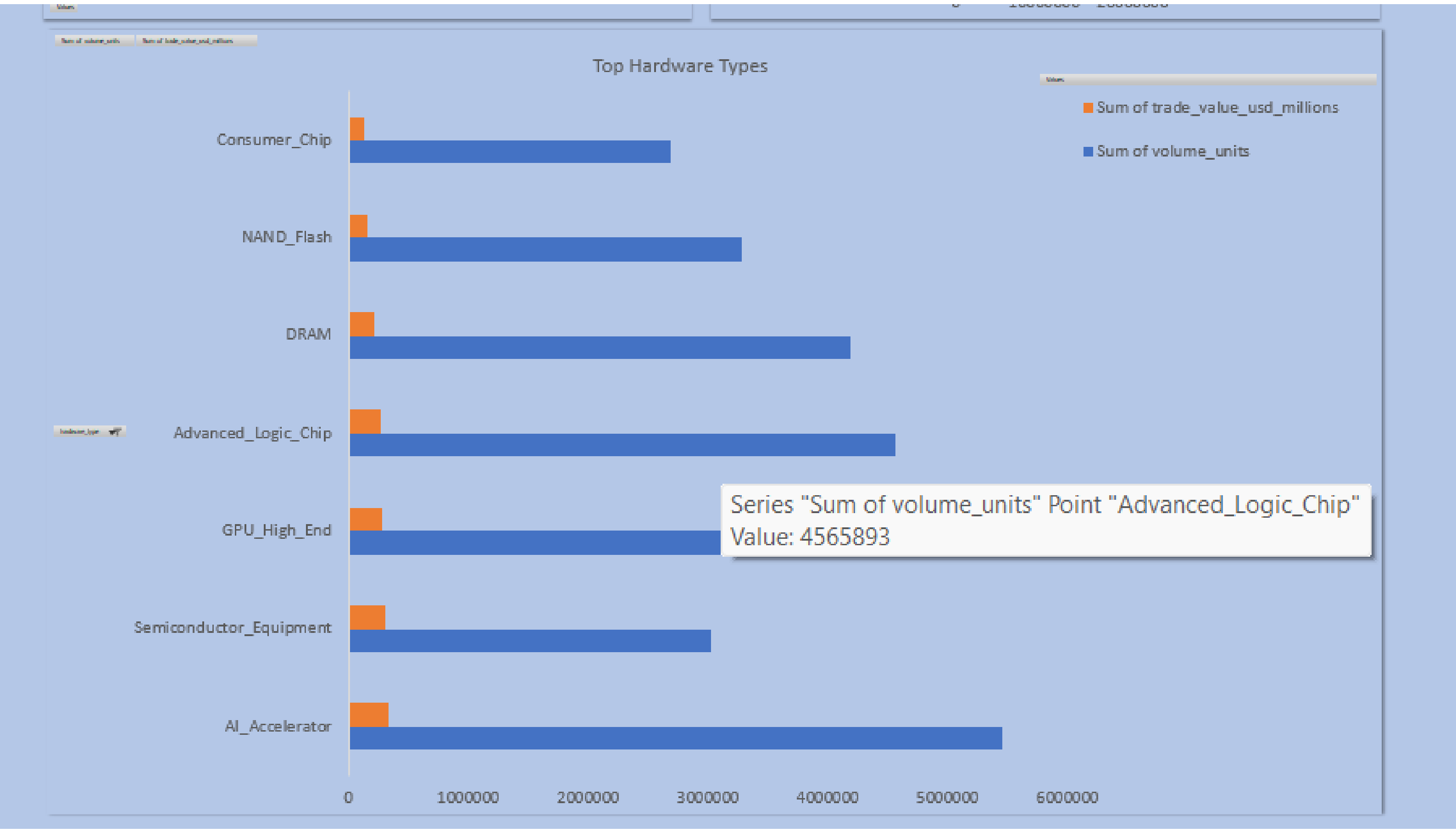

AI Semiconductor Trade Intelligence System
Executive Dashboard
Trade Value Million
$3906768.88
Total Shipment Volume
31422521
Total Exporter Countries
11
Total Importer Countries
11
Total Suppliers
40
Trade Routes
4
Top Exporter Countries
Saudi Arabia
France
Germany
South Korea
United States
Netherlands
0
500000
1000000
1500000
2000000
Amount in Million $
Region Share
Netherlands
Taiwan
United States
Japan
South Korea
China
Germany
United Arab Emirates
France
India
Saudi A
Plot Area
Chart Area
Hardware Category
Standard
Advanced
Trade Routes
Middle-East/South-Asia Corridor
Intra-Atlantic
Asia-Pacific Maritime
Trans-Pacific / Suez
0
10000000
20000000
Sum of trade_value_usd_millions
Sum of volume_units



Top Hardware Types
Sum of trade_value_usd_millions
Sum of volume_units
Consumer_Chip
NAND_Flash
DRAM
Advanced_Logic_Chip
GPU_High_End
Semiconductor_Equipment
AI_Accelerator
Series "Sum of volume_units" Point "Advanced_Logic_Chip"
Value: 4565893
0
1000000
2000000
3000000
4000000
5000000
6000000

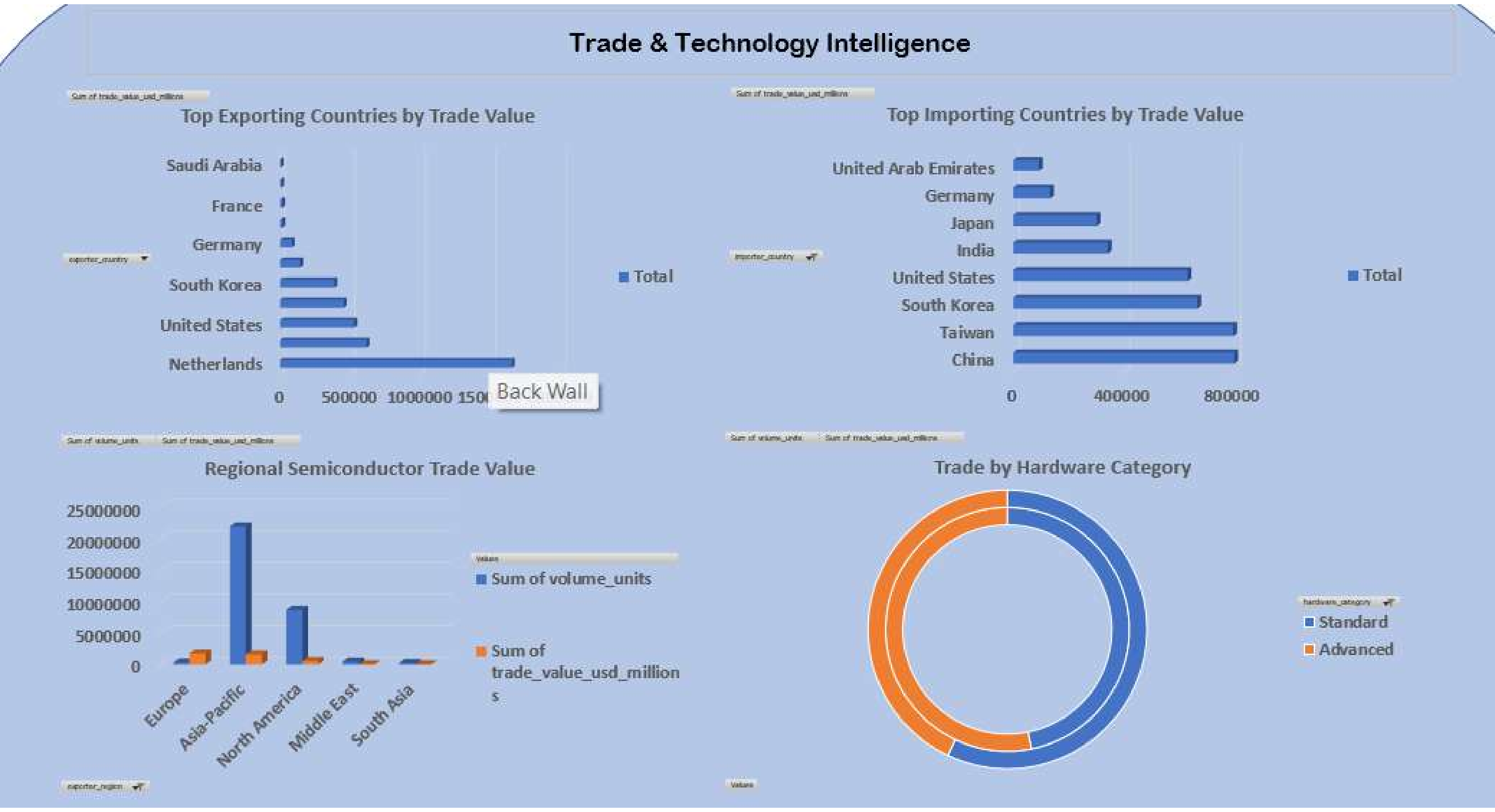
AI Executive Summary

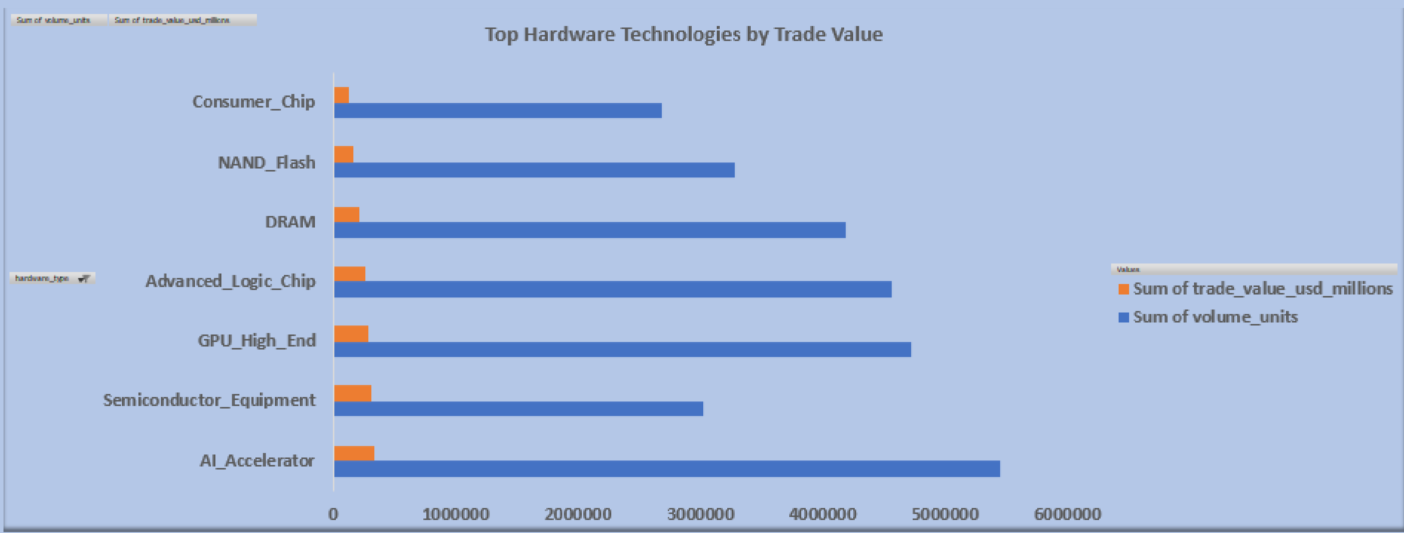
Key Findings

🌍 Semiconductor exports are highly concentrated, with **92.85%** of trade value generated by the top five exporting countries.
🛣️ **94.80%** of global trade flows through only three strategic trade routes, increasing logistics dependency.
🏭 **ASML** is the dominant supplier, while **DUV Lithography Machines** generate the highest trade value among hardware technologies.
⚠️ **Taiwan** combines significant trade value with the highest geopolitical risk, making it a critical supply chain hotspot.
🎯 Organizations should diversify sourcing, strengthen strategic partnerships with low-risk suppliers, and continuously monitor geopolitical developments to improve supply chain resilience

Executive Recommendations

🌍 **Diversify sourcing** beyond the top five exporting countries to reduce geographic dependency.
🛣️ **Develop alternative trade routes** to mitigate logistics concentration risk.
🏭 **Strengthen partnerships with ASML and other strategic suppliers** while expanding the supplier base.
⚠️ **Continuously monitor Taiwan and other high-risk regions** to manage geopolitical exposure.
💻 **Prioritize investments in lithography equipment and AI semiconductor technologies** to support future growth.

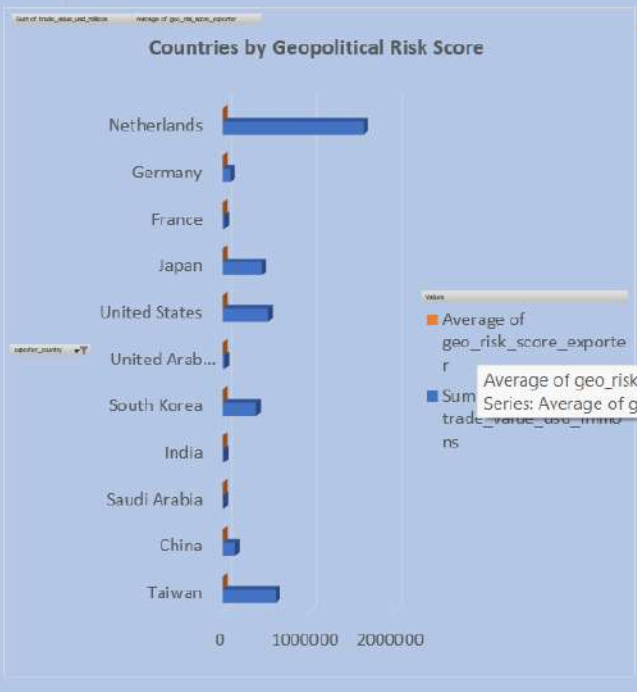

Trade & Technology Intelligence
Top Exporting Countries by Trade Value
Saudi Arabia
France
Germany
South Korea
United States
Netherlands
0
500000
1000000
Back Wall
Total
Top Importing Countries by Trade Value
United Arab Emirates
Germany
Japan
India
United States
South Korea
Taiwan
China
0
400000
800000
Total
Regional Semiconductor Trade Value
25000000
20000000
15000000
10000000
5000000
0
Europe
Asia-Pacific
North America
Middle East
South Asia
Sum of volume_units
Sum of trade_value_usd_millions
Trade by Hardware Category
Standard
Advanced

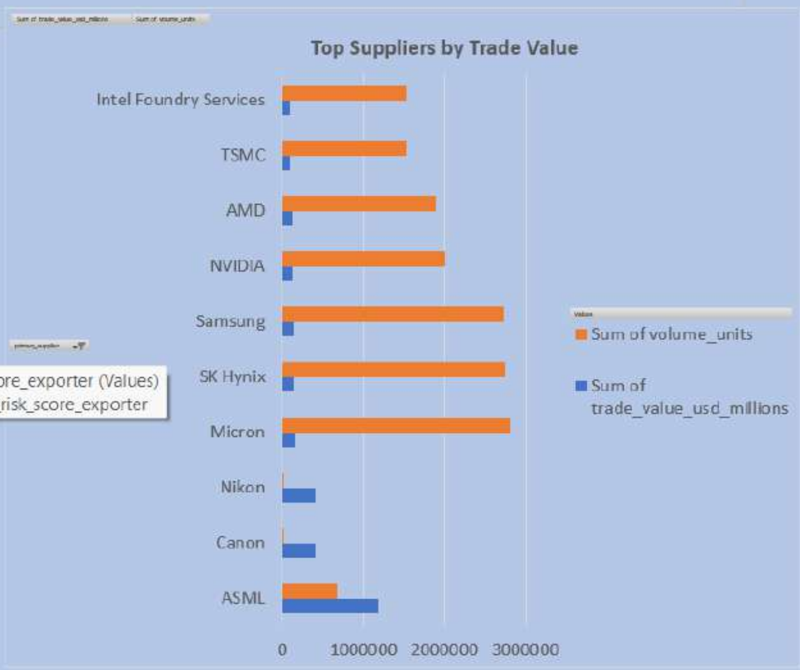


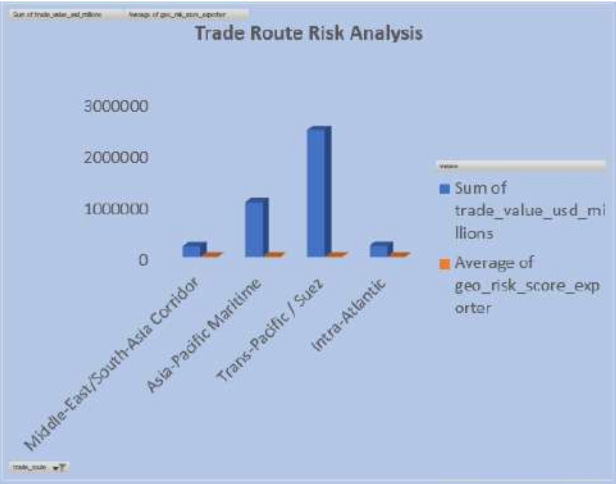
AI Trade & Technology Insights

The Netherlands dominates global semiconductor trade value, primarily through lithography equipment exports.
Taiwan specializes in advanced logic chips, reinforcing its critical role in semiconductor manufacturing.
The United States leads AI-focused semiconductor technologies, including GPUs and AI Accelerators.
💻 **Lithography equipment represents the highest-value technology segment**, highlighting its strategic importance.
📈 Global semiconductor trade remains highly concentrated among a small number of countries and advanced technologies.

Risk & Supplier Intelligence Dashboard







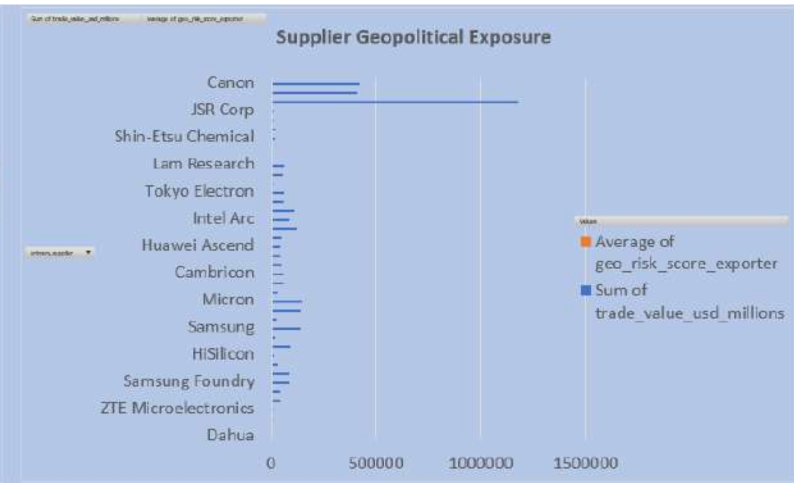


Chart Area **Supply Chain Concentration**

Top 5 Countries	Top 5 Suppliers	Top 3 Trade Routes	Top 5 Hardware
92.85%	80.43%	94.80%	82.49%

🤖 AI Risk Intelligence

• Taiwan has the highest geopolitical risk among major exporting countries.

• ASML remains the most strategically important supplier.

• Nearly 95% of semiconductor trade depends on only three logistics corridors.

• Supply chain concentration increases vulnerability to geopolitical disruptions.

• Diversifying suppliers and transportation routes can significantly improve resilience.

Executive Risk Recommendations

• Diversify sourcing beyond the top five exporting countries.

• Reduce dependency on high-risk logistics corridors.

• Strengthen partnerships with strategically important suppliers.

• Continuously monitor geopolitical developments affecting Taiwan and other high-risk regions.

• Develop contingency sourcing plans for critical semiconductor technologies.

Strategic Decision Center

Best Exporting Country= Netherlands

Highest Trade Value = $1645222.44

Highest Risk Country = Taiwan

Risk Score: 8.54

Most Critical Supplier = ASML

Highest Trade Value= $1177313.72

Most Critical Trade Route= Trans-Pacific / Suez

94.8% Trade Dependency

Critical Technology = DUV Lithography Machine

AI Growth Technology = AI Accelerator

ALERT 1

Taiwan combines
High Trade
+
Highest Geopolitical Risk

ALERT 2

94.80%
of trade depends on only
3 Trade Routes

ALERT 3

Top 5 Countries control
92.85%
of exports

ALERT 4

ASML remains
the most strategically important supplier

Metric	Status
Geographic Concentration	High
Supplier Concentration	Moderate
Route Concentration	Critical
Technology Concentration	High
Overall Supply Chain Risk	Moderate

AI Executive Insights

The semiconductor ecosystem is highly concentrated across geography, logistics, suppliers, and critical technologies. The Netherlands dominates export value, while Taiwan represents the greatest geopolitical vulnerability. ASML remains the most strategically significant supplier, and DUV Lithography Machines continue to underpin semiconductor manufacturing. Organizations should prioritize supply chain diversification, strengthen relationships with strategic suppliers, and closely monitor geopolitical developments to improve long-term resilience.

Strategic Recommendations

1. Diversify sourcing beyond the top exporting countries.

2. Reduce dependency on high-risk logistics corridors.

3. Strengthen strategic partnerships with ASML and other critical suppliers.

4. Continuously monitor geopolitical developments affecting Taiwan.

5. Increase investment in lithography and AI semiconductor technologies.

6. Establish contingency sourcing plans for critical hardware.

7. Track supplier concentration and geopolitical exposure regularly.

8. Expand partnerships across lower-risk regions to improve resilience.